

MAINS MASTER PROGRAM (MMP) 2024

ECONOMY-12; INFRASTRUCTURE-1

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2. SYLLABUS

- Infrastructure: Energy, Ports, Roads, Airports, Railways, etc.
- Investment models.

3. PYQS

1. What is the status of digitalization of Indian economy? Examine the problems faced in this regard and suggest improvements. [Mains 2023, 10 marks, 150 words]
2. Why is **Public Private Partnership (PPP)** required in infrastructure projects? Examine the role of PPP model in the redevelopment of Railway Stations in India [Mains 2022, 10 marks, 150 words]
3. Investment in **Infrastructure** is essential for more rapid and inclusive economic growth. Discuss in the light of India's experience [Mains 2021, 15 marks, 250 words]
4. "Access to affordable reliable, sustainable and **modern energy** is the sine qua non to achieve Sustainable Development Goals (SDGs). Comment on the progress made in India in this regard [Mains 2018, 10 marks, 150 words]
5. Examine the development of **airports** in India through joint ventures under **Public-Private Partnership (PPP)** model. What are the challenges faced by the authorities in this regard [Mains 2017, 10 marks, 150 words]
6. What are '**Smart Cities**'? Examine the relevance for urban development in India. Will it increase rural-urban differences? Give arguments for 'Smart Villages' in the light of PURA and RURBAN mission [Mains 2016, 12.5 marks, 250 words]
7. **National Urban Transport Policy** emphasizes on 'moving people' instead of 'moving vehicles'. Discuss critically the successes of the various strategies of the Government in this regard. [2014, 12.5 marks, 200 words]
8. Explain how **PPP arrangements**, in long gestation infrastructure projects, can transfer sustainable liabilities to the future. What arrangements need to be put in place to ensure that successive generations' capacities are not compromised? [2014, 12.5 marks, 250 words]
9. Adoption of **PPP model** for infrastructure development of the country has not been free of criticism. Critically discuss pros and cons of the model [2013, 12.5 marks, 200 words]
10. Write a note on India's **Green Energy Corridor** to alleviate the problem of conventional energy [2013, 10 marks, 150 words]
11. Are the '**Dedicated Freight Corridor**' railway project and the '**Golden Quadrilateral**' road project mutually complementary or competitive? Assess [10 marks, 150 words]

4. SIGNIFICANCE OF INFRASTRUCTURE SECTOR

- **Practice Questions:**
 - » **Question 1:** Investment in Infrastructure is crucial for inclusive and long term development of the country. Discuss the key recent initiatives by GoI to promote sustainable Infrastructure Development in India [10 marks, 150 words]
 - » **Question 2:** Investment in **Infrastructure** is essential for more rapid and inclusive economic growth. Discuss in the light of India's experience [Mains 2021, 15 marks, 250 words]
- **Introduction:**
 - » Infrastructure is the backbone of any economy. The extent and quality of infrastructure determines the ability of the country to utilize its comparative advantage and enables cost competitiveness.
- **Investment in infrastructure is crucial for inclusive and long-term development** because:
 - » **Investment in Infrastructure is crucial for Rapid Economic Growth:**
 - Infrastructure services like electricity, telecommunications, roads, rail, ports, and airports - are critical to the development of a strong, modern, productive and efficient economy.
 - **Development of infrastructure** -> Creates jobs for engineers, designers, contractors, laborers etc.
 - **Attracts investment** -> good quality infrastructure is one of the pre-requisites for investment for both domestic and international market, which further leads to economic growth, job opportunities etc.
 - **Reduces logistic cost** -> Good quality roads, railways, custom infrastructure reduces both time and cost of logistics making our goods more competitive in the global market.
 - **Overcoming slowdown** -> For e.g. during pandemic driven slowdown, government responded by increasing capital expenditure aimed at high quality physical and social infrastructure.
 - » **Improved infrastructure can lead to inclusive growth:**
 - **Bridging regional disparities:** Investment in infrastructure is the fastest way to bridge regional disparities and attract businesses and industries to underserved region.
 - **Improved rural development** is possible through investment in infrastructure. Farmer can access both global and domestic market easily; agri-waste could be reduced.
 - » **National integration and regional development** is ensured by better connectivity through railways/airways/roadways etc.
 - » **Better social development** also improves with infrastructure -> for e.g. better telecom infrastructure can improve health and education facilities through telemedicine and Massive Open Online Courses (MOOCs).
 - » **Reduces the scope of radicalization and left wing extremism** -> Extremist ideas take hold in those people easily who feel isolated and alienated.
- **Keeping the significance of infrastructure in mind**, government has launched several initiatives, specially after the 1991 LPG reforms to improve the infrastructure situation in the country.
 - » **Increasing Investment:**

- **High Capital allocation for infrastructure:** In last few years, government's capital expenditure has regularly increased and the budgetary allocation for the FY25 for the same has reached Rs 11.11 lakh crores (3.4% of the GDP)
 - **National Infrastructure Pipeline (NIP)** aims to mobilize more than 100 lakh crore rupees between 2020-25 for various infrastructure projects.
 - **National Monetization Pipeline** will also contribute to mobilization of resources for infra projects through monetization of various brown field government projects.
 - **National Bank for Financing, Infrastructure and Development (NaBFID)** has been set up as developmental financial institution to set in motion a virtuous investment cycle.
 - **Project Development Cells (PDCs)** have been set up in all concerned ministries and departments of GOI to provide an institutional mechanism to fast track investments.
- » To **enhance efficiency and cost competitiveness** - PM Gati Shakti and National Logistics Policy has been launched.
 - **PM GatiShakti** is aimed at **breaking departmental silos** and bring **more holistic and integrated planning and execution of projects**.
 - **National Logistics Policy (NLP)** aims to reduce the logistics cost in India and bring it on par with developed nations.
- » **Sector Specific Initiatives:** Other than these the traditional focus on infrastructure efforts like roadways, railways, airways, ports etc have continued with various initiatives like Bharatmala Pariyojna, Sagarmala Pariyojna, UDAN initiatives, etc.
- **Key challenges to infrastructure development in India:**
 - » **Limited Private sector participation:** Issues like regulatory uncertainties, land acquisition issues etc have hindered private sector participation in the infrastructure sector.
 - **Land Acquisition** specially has become a challenge specially after the 2013 land acquisition act.
 - » **Human Resource and Technical Shortage:** Inadequate institutional capacity, technical expertise, and human resources hinder advanced infra projects.
 - » **Ensuring Equitable Access and Inclusive Development:** The underdeveloped regions of the country are not able to invest in or attract large infra projects.
 - » **Issues related to environmental clearance:** Red tapism, complex procedures have led to delays in environmental clearance for infrastructure projects.
- **Conclusion**
 - » These initiatives can collectively bring out untapped potential of underutilized modes of infrastructure. This buffet of reform on all aspects related to infrastructure investment and project execution not just ensure economic growth and economic efficiency but also generates confidence among various stakeholders in the economy towards government functioning. IMF has observed in case of India, increased government spending on infrastructure also leads to crowding in of private infrastructure.

5. INFRASTRUCTURE FINANCING: THE PUBLIC EXPENDITURE PUSH

- Though the budgetary capital expenditure can't be equated to infrastructure spending, the infrastructure thrust of the government has led to unprecedented increase in capital expenditure.
 - » The capital expenditure of the union government increased by **2.2 times** from FY21 to FY24, while that of the state government increased by **2.1 times** during this period.
 - The share of gross budgetary support to two key connectivity segments, i.e., Railways and National Highway Authority of India (NHAI), in the total capital expenditure increased from 36.1% in FY21 to 42.9% in FY24.
 - Other than this, government has also worked towards reducing the borrowing cost of the two major infra-CPSEs - NHAI and IRFC.
 - » In **FY25**, as per the budget estimate, central government plans to spend **Rs 11.11 lakh crores (3.4% of the GDP)** on capital expenditure.
 - » The support of Union Government for capital expenditure of the state government and institutions increased by 31.6% during FY21 and FY24.
- **Issue:** The recent infrastructure thrust in India, especially the surge in connectivity projects, has banked predominantly on public expenditure.
 - » **Bank Credit for infrastructure** (between March 2023 to March 2024) was just **Rs 79,000 crores**, much less than government budget allocation.
 - **Bank Credit** has also been concentrated in few sectors - roads, airports, and power.
 - » **Gross inflow of External Commercial Borrowing (EBC) for infrastructure** was **USD 9.05 billion** in FY24.
 - » **Debt and Equity** issuances in the capital market was also low (just Rs 1 lakh during FY24)
 - » **Real Estate Investment Trust (REIT)** have raised **Rs 18,840 crores** from year 2019 to 2024, while **Infrastructure Investment Trusts (InvITs)** raised a total of **Rs 1,11,294 crores** in the last five years (2019-2024).
- **Solution:** Need to mobilize private investment in Infrastructure through PPP

6. PUBLIC PRIVATE PARTNERSHIP (PPP)

- **Practice Questions:**
 1. Why is **Public Private Partnership (PPP)** required in infrastructure projects? Examine the role of PPP model in the redevelopment of Railway Stations in India [Mains 2022, 10 marks, 150 words]
 2. Explain how **PPP arrangements**, in long gestation infrastructure projects, can transfer sustainable liabilities to the future. What arrangements need to be put in place to ensure that successive generations' capacities are not compromised? [2014, 12.5 marks, 250 words]
 3. Adoption of **PPP model** for infrastructure development of the country has not been free of criticism. Critically discuss pros and cons of the model [2013, 12.5 marks, 200 words]
- **Introduction:**
 - » In 1997, the report of the **Rakesh Mohan Committee (RMC)** concluded that India's problem was that of **poor infrastructure holding back development**. It highlighted how bringing in the private sector into most areas of infrastructure can reduce this problem.
 - » This private investment in infrastructure has mainly come in the form of PPPs in India.

- » **Public Private Partnership (PPP)** is a collaborative arrangement between government and private sector to jointly plan, mobilize resources, develop, and/or operate infrastructure projects.
 - According to the Canadian Council for Public Private Partnership PPP is defined as "Co-operative venture between the public and private sectors, built on the expertise of each partner that best meets clearly defined public needs through appropriate allocation of resources, risks and rewards".

- **Significance of Public Private Partnerships:**

- **Mobilization of Resources:**(For e.g. in the BOT (Toll/ Annuity) model of road construction, private player invests the entire initial money for the construction of the road project).
 - **Getting Private Sector expertise and Innovation:** E.g. in the EPC model, private sector engineers and construct the entire road.
 - **Risk Sharing:** Long gestation period of infra projects bring along its own risks due to uncertain revenue stream in long run. PPP projects allows government to share these risks with private players.
 - **Increased Efficiency and Reduced cost of the project:** Private sector comes along with efficiency and reduced scope for corruption - this leads to timely and economic completion of various infra projects.
 - **Increased Transparency and Accountability:** PPP projects are subject to public scrutiny and the private players are also accountable to their shareholders.
 - **Better Infrastructure:** All the above advantages eventually contribute to better quality and quantity of infrastructure for general public.
- But, the **success of PPPs** lies in the robustness of institutional structure, financial support, and use and availability of standardized documents, such as Model Requests for Qualifications (RFQ), Model Request for Proposal (RFP) and Model Concession Agreement (MCAs).
- Government of India has taken several measures:
- » **Public Private Partnership Appraisal Committee** - has been set up in 2006 to act as the apex body for appraisal of central sector PPP projects.
 - Between FY15 to FY24, it has recommended 77 projects with a total cost of Rs 2.4 lakh crore.
 - » **Viability Gap Funding (VGF) Scheme, 2006:** It provides financial assistance to financially unviable but socially/economically desirable PPP projects.
 - **Economic Sector Projects** may get upto 40% of Capex as VGF grant.
 - **Social Sector Projects** include higher provisions of VGF grant. It may get upto 80% of CAPEX and upto 50% of the Operating Expenditure (OPEX).
 - » **India Infrastructure Project Development Fund (IIPDF) Scheme** notified in Nov 2022
 - It aims to develop quality PPP projects by providing necessary funding support to project sponsoring authorities, both in the central and the state governments, for creating a shelf of bankable and viable PPP projects by on-boarding transaction advisors.
 - » **Reference guides** for setting up state PPP Units, PPP project appraisal, and project implementation mode selection have been made.
 - » **Web-based toolkits, post-award contract management toolkit, and contingent liability for project sponsoring authorities** have been developed to help them in PPP structuring.

- **Several types of PPP Models** are used in India in different sectors:
 - **EPC Model (Engineering, Procurement, and Construction):** In this model, the cost of project is completely borne by government. Private sector with its expertise is responsible for engineering, procuring raw material and constructing the project. Ownership remains with government.
 - **Built Operate and Transfer (BOT) model** involves private player entity designing, financing, constructing, operating, and maintaining an infra projects for a specific period. After the specified period, the ownership is transferred back to government. This model has been used in sectors like Roadways, Ports, Airports and Power Generation. It can be of two types - BOT (Toll) & BOT (Annuity).
 - **Hybrid Annuity Model (HAM):** It is a mix of EPC and BOT (Annuity) model.
 - **Build Own Operate (BOO):** The private sector entity builds and owns the asset, and then operates it for a specified period of time.
 - Government has agreed to "buy" the goods and services delivered by the project on mutually acceptable terms and circumstances.
 - **Build Own Operate Transfer (BOOT):**
 - It is a model of PPP in which a private company is granted a concession to finance, build, own, and operate a project for a specified period of time. At the end of the concession period, the project is transferred back to government.
 - E.g. of project under BOOT model, Delhi Mumbai Expressway, The Mumbai Metro, the Bangalore International airport etc.
 - It involves a private sector entity being responsible for the complete lifecycle of the project, including design, financing, construction, operation, and maintenance. However, here private sector entity retains ownership of the project even after the concession period.
 - **Build Own Lease Transfer (BOLT):** It is a PPP model in which a private company is granted a concession to finance, build, own and lease a project to the government for the specified period of time. At the end of the concession period, the project is transferred back to government. Some notable BLT projects in India are, the Delhi-Gurgaon Expressway and the Mumbai-Pune Expressway.
 - **Design Build Finance and Operate (DBFO):** It allows a private sector to design, build, finance, and operate a project for a specified period of time. This public sector client retains the ownership of the project, but the private sector contractor is responsible for all aspects of its delivery.
 - E.g. Delhi Metro: Project was awarded to a consortium of private companies, which designed, built, financed, and operated the metro for a period of 30 years. At the end of the concession period, the metro will be transferred back to government.
 - **Lease Developed Operate (LDO) Model:** Private company is granted a concession to finance, develop, and operate a project for a specified period of time. The government sector retains the ownership, but the private sector is responsible for all aspects of its delivery. At the end of the concession period, government may choose to operate the project itself, or it may contract with another private company to operate the project.
 - **Rehabilitate-Operate-Transfer (ROT) Model:** Under this model, government allows private promoters to rehabilitate and operate a facility during a concession period. After the concession period, the project is transferred back to government / local bodies.

- **Challenges of PPP Model:**

- **Institutional Inadequacies:** An overly regulated institutional and legal framework with issues such as complexity, fragmentation, and lack of accountability results in inefficient implementation of PPP Projects.
- **Lack of expertise in identifying projects, forming contract, choosing the right PPP models etc.**
 - This leads to difficulty in assessing the investment needs and the duration required to improve service and operational efficiencies, which is crucial to the success of PPP.
 - **It leads to projects being stuck due to disputes in existing contracts.**
- **Lack of Transparency and Accountability:** There is inadequate information on PPPs in public domain. Private companies often don't share data in the name of business secrecy.
- **Profit Motive as opposed to Public Service:**
 - **Service efficiency** in the delivery of PPP models is lacking due to profit motive.
 - **Crony capitalism** has also emerged as a major challenge in several sectors.
- **Communication and Stakeholder engagement** is generally weak and lacking in the projects.
- **Lack of balanced assessment and treatment of Risk Sharing** in the PPP projects.
- Issues such as non-availability of capital and land acquisition challenges also hinder PPP projects.
- **Long Gestation period and delays** in the implementation has led to a lot of loans becoming Non-Performing Assets (NPAs).
- **Uncertainty:** The regulatory and legal framework of PPP models in India are still evolving. This can lead to uncertainty for investors.

- **Recommendations of Vijay Kelkar Committee:**

- Vijay Kelkar Committee on "Revisiting & Revitalizing the PPP model of infrastructure Development" was set up in the Union Budget of FY15-16. It recommended:
 - The Need of PPP contract to be **more focused on service delivery**.
 - The need to identify, balance and allocate risks amongst the different stakeholders.
 - Viability Gap Funding for unviable social and economic projects
 - Careful monitoring of performance as well as managing the risk.
- **Other suggestions:**
 - **Strengthening Contracts:** PPP model contracts should allow flexibility in resetting the targets or renegotiating funding allocations
 - **Strengthening Institutional Capacities:** Bringing experts from the area of contract formation, risk assessment etc. Setting up a PPP adjudication tribunal for better dispute redressal.

- **Conclusion:**

- The success of PPP model in India will depend on the ability of the government to address these factors and to create an environment that is conducive to private investment in infrastructure.

7. NATIONAL INFRASTRUCTURE PIPELINE (NIP)

- **Example Questions:**

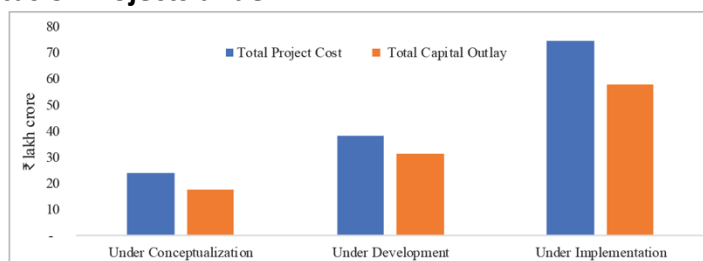
- What is National Infrastructure Pipeline? What is its significance for India's development [10 marks, 150 words]

- In 2020, Ministry of Finance estimates that to achieve a GDP of \$5 trillion by 2024-25, India needs to spend about \$1.5 trillion (Rs 111 lakh crore) over these years in infrastructure. Keeping this in mind, government has launched National Infrastructure Pipeline (2020-25) with projected infrastructure investment of around Rs 111 lakh crores. It also envisages to improve project preparation and attract investment, both domestic and foreign in infrastructure.

- Progress So Far:

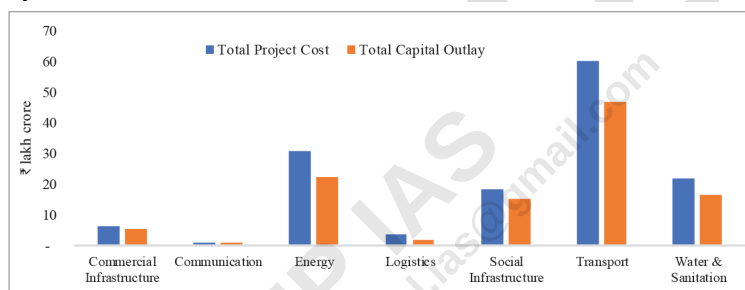
- » The NIP currently has 9,288 projects with a total investment of more than Rs 108 lakh crore under different stages of implementation.

» Status of Projects under NIP



Source: Department of Economic Affairs.
Note: Data as of 13 January 2023

» Transportation Sector Dominates the NIP



Source: Department of Economic Affairs.
Note: Data as of 13 January 2023

» Jointly funded by Central Government, State Government and Private Sector

- Some steps to improve the implementation of NIP:

» Invest India Grid (IIG):

- NIP is hosted on the Invest India Grid (IIG) platform and provides and provides opportunities states/UT and ministries to collate all major infrastructure projects at a single location. It is thus a centralized portal to track and review project progress across all economic and social infrastructure sub-sectors. It also provides the project sponsoring authorities to showcase investment opportunities to national and international investors.

» Project Monitoring Group (PMG)

- It is an institutional mechanism put in place by the government for resolution of issues related to large-scale projects. It is also involved in fast tracking of approvals/clearances for projects with an anticipated investment of Rs 500 crore and above. Now it has been proposed to integrate NIP and PMG portals. PMG portal will pick up data, as per requirements (project cost of Rs 500 crore or more), from the NIP database. This will

save substantial time and effort by Ministries and States/UTs and ease monitoring of large-scale projects.

- **Significance:**

- **Economic Development:** Well planed NIP projects will enable more infra projects, grow businesses, and create more jobs. It will also promote ease of doing business and enhance export competitiveness of Indian products.
- **Inclusive Growth:** Improved infrastructure will improve ease of living, and will provide equitable access to infrastructure for all making growth more inclusive.
- **Government:** Enhanced economic activities will lead to creation of additional fiscal space by improve revenue base of government. It also ensures quality of expenditure focused on productive areas.
- **Advantages for Developers:** NIP provides a better view of project supply, provides time to be better prepared for project building, reduces aggressive bids/failures in project delivery and ensure enhanced access to source of finance as a result of increased investor confidence.
- **Banks/Financial Institutions/Investors:** Confidence in the identified projects are likely to be higher and they are less likely to suffer stress given active project monitoring, thereby less likely of NPAs.

- **Way Forward:**

- » **Improving Project Preparation Process** to avoid delays and reduce cost. The **preparation framework** shall consist of a transparent policy and legislative framework, an empowered public institution for infrastructure planning, multi-stage reviews, audits and approvals for quality assurance etc.
- » **Enhancing execution capacity of private sector participants** - There is a need of deep pool of experienced developers with required competence and execution capacity. Here collaboration and joint ventures with strong global infrastructure developers can be crucial.
- » **Robust Enabling Environment** through appropriate policy framework and a well developed public institutional capacity.
- » **Institutionalization and Efficiency of Dispute Resolution**
 - **Steps already taken:** The Commercial Courts Act 2015, The Specific Relief (Amendment) Act 2018 and the New Delhi Arbitration Centre Act 2019
 - **What more can be done?** Ministry level committees to resolve complex contractual disputes as mediation mechanism that can settle disputes out of courts.
- » **Strengthening infrastructure quality** - Creating a National Framework for Infrastructure Quality for each sector based on global and national standards.
- » **Financial Sector Reforms**
 - **Revitalizing the bond and credit market**
 - **Strengthening the Municipal Bond Market in India**
 - **Revitalizing Asset Monetization**
 - **Enabling User Charges to finance infrastructure**
 - **Long term financing Landscape**

- **Conclusion:**

- » Overall, the National Infrastructure Pipeline (NIP) is a transformative initiative that aims to propel India's economic growth, improve living standards and position country as a global powerhouse.

8. NATIONAL MONETIZATION PIPELINE (NMP)

- **Practice Questions:**
 - » Discuss the objective and significance of the National Monetization Pipeline in the context of India's economic development [10 marks, 150 words]
- **Background**
 - » **Asset Monetization** is one of the key recommendations of National Infrastructure Pipeline (2020-25). Consequently, National Monetization Pipeline was announced in 2021. It focuses on the principle of 'asset creation through monetization' and thus taps private sector investment for new infrastructure creation.
- **Details:**
 - » Asset monetization entails a **limited period license/lease of a brownfield underutilized asset** owned by government or a public agency, **to a private sector entity** for an **upfront or periodic consideration**.
 - The private sector entity is expected to **operate and maintain** the asset based on the terms of the contract/concession, generating returns through higher operating efficiencies and enhanced user experience.
 - **Public authority**, which receives the fund, will invest it in new infrastructure or deploy it for other public purposes.
 - » A **robust asset pipeline** has been prepared to **provide a comprehensive view to investors and developers** of the investment avenues in infrastructure.
 - It includes selection of de-risked and brownfield assets with stable revenue generation profile (or long rights) which will make for an attractive investment option.
 - » Total **indicative value of NMP** for core assets of the Central Government has been **estimated at Rs 6.0 lakh crore** over 4-year period (FY22 - 25) (5.4% of the total infrastructure investment envisaged under NIP)
 - » **National Land Monetization Corporation (NLMC):** Cabinet approved the setting up of the NLMC to monetize surplus land and building assets of CPSEs and other agencies linked to government (March 2022)
- **Progress So far: ESI 2023-24:**
 - » Ministries are proactively working on developing a pipeline and transactions have been undertaken in line with their strategic investments.
 - In FY22 and FY23, transactions aggregating to about **Rs 2.3 lakh crore** in accruals or private investments were completed.
 - In FY24, transactions aggregating to **Rs 1.51 lakh crore** in accruals or private investments were completed, 1.55 times those achieved in 2021-22.
- **Significance: Positives**
 - » **Infrastructure Upgradation and Development:** The NMP aims to enhance infrastructure development in the country by leveraging the potential of existing assets.

- » **Unlocking of Capital and Revenue Generation:** The funds generated can be used for financing new infrastructure projects without increasing the fiscal deficit.
- » **Efficient Asset Management: Promotes the use of complete assets** which are languishing or it is not fully monetized or is under utilized.
- » **Job Creation and Skill Development:** It will attract investment, promote entrepreneurship, and stimulate economic growth.
- » Clear framework for monetization and given **potential investors a ready list of assets** to generate investment interest.
- » Along with disinvestment, monetization of idle land is **part of the Centre's strategy to reduce its business presence to a bare minimum** and generate resources for future asset creation.
- **Key challenges:**
 - » **Valuation of Assets** is a complex process. There may be challenges in accurately assessing the future revenue, depreciation factors etc.
 - » **Lack of government's expertise in designing long term contracts:**
 - Contractual disputes can pose hurdle in implementation process.
 - » **Monetizing assets with low revenue potential may be a challenge**
 - » **Asset Management and Quality Maintenance** - Poor monitoring may lead to poor quality of asset management.
 - » **Sector Specific Challenges:**
 - **Level of capacity utilization in gas and petroleum pipeline networks** -> lack of capacity utilization has been due to lower gas supplies
 - **Regulated tariffs in power sector assets**
 - **Low interest of investors in national highways** of less than 4 lane.
- **Other limitations:**
 - » **Threat of monopolization:** Because of limited number of private players, bidding for assets may not be totally fair.
 - » **Public welfare** may get ignored (which should be the basic focus of public sector assets)
 - » **Why are the assets being poorly managed** at the first place. If its poor management, Asset Monetization makes sense. But if the issues are structural, even asset monetization will not solve the problem.
- **Way Forward:**
 - » **Set up a centralized institution with the skills and responsibility to oversee contract design, bidding and implementation,** separate from, but with appropriate assistance of, the concerned line ministries.
 - » **Robust dispute resolution mechanism**
 - » Government should start with sectors that offer greatest cash flow predictability and the least regulatory uncertainty before expanding the experiment.
 - » Ensure that resources raised from the NMP are used to fund new asset creation.
- **Conclusion:**
 - » While the ideal of monetization seems great, execution of the plan remains key to its success.

- Need of PM GatiShakti:

- » There are many infrastructure projects like roadways, railways, airways, waterways, internet connectivity (optical fiber), Gas Pipelines etc. These projects come under different ministries, leading to lack of coordination in planning and implementation of these projects. This leads to duplication of work, delays, financial loss, increase in cost etc., which eventually puts more burden on the public exchequer and impacts the quality of services reaching people.
 - E.g. 1: Newly built roads being dug by water departments to lay pipelines.
 - E.g. 2: Newly built fertilizer factory not working properly as the gas supply infrastructure isn't available.
 - E.g. 3: Separate tunnel for roadways and railways

- Details

- » **PM GatiShakti** is aimed at **breaking departmental silos** and bring **more holistic and integrated planning and execution of projects** with a view to address the issue of Multi-Modal connectivity and last mile connectivity.
 - This will help in **bringing down the logistic cost** and will translate into enormous financial gains to consumers, farmers, youth as well as those engaged in businesses.
- » The PM GatiShakti National Master Plan entails creation of a common umbrella platform with all infrastructure projects pertaining to various ministries/ departments **incorporated within a comprehensive database** for efficient planning and implementation on a real-time basis.
 - The projects pertaining to **seven engines (roads, railways, airports, ports, mass transport, waterways, and logistic infrastructure)** in the NIP have been aligned with PM GatiShakti framework.
 - The portal will also **allow various government departments to track, in real time and at one centralized place, the progress of various projects**, especially those with multi-sectoral and multi-regional impact. This gives every department/ministry a visibility into other department/ministry's projects. This will enable various government departments to synchronize their efforts into a multi-modal network.
 - The portal will also **highlight all the clearances** any new project would need, based on its location - and allow stakeholders to apply for these clearances from the relevant authority directly on the portal.
- » **Six Pillars of PM Gati Shakti:** Comprehensiveness, Prioritization, Optimization, Synchronization, Analytics and Dynamic.
- » **How were inter-ministerial issues resolved earlier?**
 - At regular meetings of infrastructure related ministries.
 - PM PRAGATI (Pro-Active Governance and Timely Implementation) portal also helped in resolution of several issues even prior to inter-ministerial meetings.
 - **How would Gatishakti portal help?**
 - It will reduce the human intervention required as ministries will be in constant touch, and projects will be reviewed by the project monitoring group in **real time**.
- » **Progress as of March 2024:**
 - **43 ministries** have been onboarded on PMGS-NMP portal.
 - **1,530 data layers** (642 ministry data layers & 888 state data layers) of Ministries and states have been uploaded on the portal.

- **16 ministries** have dedicated PMGS Cell which has streamlined project planning.
 - **Network Planning Group**, a central-level institutional mechanism of the PMGS-NMP has evaluated 149 project proposals with total estimated project cost of Rs 13.3 lakh crore.
- **Central Universities (Amendment) Act, 2022** - by Ministry of Education - has converted the National Rail and Transportation University into the Gati Shakti Vishwavidyalaya (GSV) - an autonomous central university. Till now it was functioning as a deemed university.
- It will address the need of talent in transportation sector.
- **Significance:**
- This will help raise productivity, and accelerate economic growth and development
 - Discuss various significance of Infrastructure Development